Notes: Lerner & Tirole (AER, 2006)
A Model of Forum Shopping

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1 Big picture

- **Question.** When an owner or sponsor needs an intermediary to certify an idea, technology, security, or standard, which certifier does she choose, and is that choice socially desirable?
- **Core setting.** A sponsor chooses a certifier before users know the true value of the proposal. Certifiers differ in how friendly they are to the sponsor versus how protective they are of users.
- **Forum shopping.** The sponsor searches for the most sponsor-friendly certifier that is still credible enough for users to adopt the proposal after endorsement.
- **Design concessions.** The sponsor may also choose concessions to users, such as lower royalties, more compatibility, disclosure of intellectual property, or other design improvements.
- **Main result.** Stronger proposals choose less stringent, more sponsor-friendly certifiers and make fewer concessions. Weaker proposals need more credible certifiers and more concessions.
- **Social result.** For attractive proposals, private forum shopping is too sponsor-friendly: the sponsor picks an excessively complacent certifier and offers too few concessions. For weak proposals, the private choice may already be constrained by user participation and regulation may not improve welfare.
- **Extensions.** The paper studies property morphing, sponsor downstream presence, private information, certifier market power, multiple user groups, and competition among rival sponsors.
- **Competition result.** Competition among rival sponsors makes sponsors use less friendly, more credible certifiers than monopoly sponsors, and may lead to single, differentiated, or dual submissions.
- **Takeaway.** Certification is a strategic market-design problem. A certifier's credibility, bias, and ability to demand concessions jointly shape adoption, welfare, and strategic competition.

2 Incremental contribution of the paper

- **Strategic choice of certifier.** Prior certification models often treat the certifier as fixed. Lerner and Tirole make the choice of certifier endogenous and analyze how the sponsor shops among possible forums.
- **Certifier bias as an allocative instrument.** The paper formalizes certifier friendliness with a parameter that weights sponsor profit relative to user surplus. Certification is not just information transmission; it embeds governance and preference weights.
- **Joint certification and design.** The model links the choice of certifier to the choice of concessions or design modifications. The sponsor uses both tools to make adoption possible while preserving profit.
- **Private versus social forum shopping.** The paper identifies when the sponsor's chosen forum is too complacent from a welfare perspective and when regulation of only one dimension can backfire.
- **Multiple-user and network-externality logic.** The paper explains why centralized certification can dominate multiple parallel certifications when different user groups must coordinate on adoption.
- **Competition among sponsors.** The paper adds rival proposals and shows that competition can push sponsors toward tougher certification, differentiated strategies, or dual submissions.
- **Applications across economics.** Although the motivating application is standard-setting organizations, the theory applies to journals, underwriters, auditors, rating agencies, venture capitalists, and other certifiers.

Interpretation: The incremental contribution is to treat certification as a strategic institutional-choice problem. The paper asks not merely whether a certifier is informative, but why a sponsor chooses one certifier rather than another and how that choice changes user welfare.

3 Data and measurement / model primitives

This is a theoretical paper. Its "data" are model primitives: user benefit, sponsor profit, certifier bias, concessions, adoption thresholds, and the distribution of unknown quality.

3.1 Owner, users, and proposal quality

The sponsor owns an idea, property, technology, or standard. If users adopt it, the sponsor earns profit $\pi(c)$, where c is a concession or design improvement favorable to users. Concessions are costly to the sponsor:

$$\pi'(c) < 0, \qquad \pi''(c) \leq 0.$$

Users obtain utility

$$U = a + b + c.$$

Here a is common-knowledge attractiveness, b is unknown user benefit, and c is the concession.

Interpretation: The unknown b creates the need for certification. Users will adopt only if the endorsement gives them sufficiently favorable beliefs about b.

3.2 Certifiers and friendliness

A certifier is indexed by α. A larger α means the certifier places relatively more weight on sponsor profit and is therefore more sponsor-friendly. A certifier endorses when

$$a + b + c + \alpha\pi(c) \geq 0.$$

This creates an endorsement threshold

$$b \geq b^*(\alpha, c) = -[a + c + \alpha\pi(c)].$$

Interpretation: A friendly certifier has a lower threshold and endorses more often. A tough certifier has a higher threshold and is more credible to users.

3.3 User beliefs after endorsement

Let

$$m(b) \equiv E[\tilde{b} \mid \tilde{b} \geq b]$$

be the conditional expected benefit after observing an endorsement by a certifier with cutoff b. Users adopt after endorsement iff

$$a + m(b^*) + c \geq 0. \tag{1}$$

Interpretation: Credibility requires that endorsement be sufficiently selective. If the certifier is too friendly, an endorsement carries little information and users reject.

3.4 Forum-shopping object

The sponsor chooses both a forum and possibly a concession. For a given concession, the sponsor chooses the most friendly certifier consistent with user adoption. The lowest cutoff that induces adoption is

$$b^* = m^{-1}[-(a + c)]. \tag{2}$$

The corresponding maximum credible friendliness satisfies

$$m(b^*) - b^* = \alpha\pi(c). \tag{3}$$

Interpretation: Equations (2) and (3) summarize forum shopping. The sponsor wants the lowest possible cutoff, but cannot choose one so low that users ignore the endorsement.

4 Models and theoretical specifications

4.1 Baseline owner problem

If the design/concession is endogenous, the sponsor solves

$$\max_c [1 - F(m^{-1}[-(a+c)])]\pi(c).$$

The first-order condition can be written as

$$\frac{1}{m(b^*) - b^*} + \frac{\pi'(c)}{\pi(c)} = 0, \tag{4}$$

with $b^* = m^{-1}[-(a+c)]$. Combining the credibility condition with the design condition yields

$$1 + \alpha\pi'(c) = 0. \tag{5}$$

Interpretation: The sponsor balances a higher probability of endorsement against the profit cost of concessions. A weaker proposal needs larger concessions or a tougher certifier to be believed by users.

4.2 Private and social choices

Let w denote the social weight on sponsor profit in the planner's objective. A planner would choose concessions and adoption to maximize weighted social surplus:

$$E[X(b)(a + b + c + w\pi(c))]$$

subject to user participation:

$$E[X(b)(a + b + c)] \geq 0.$$

In the unconstrained social optimum, the planner would choose

$$1 + w\pi'(\hat{c}) = 0$$

and adoption whenever

$$a + b + \hat{c} + w\pi(\hat{c}) \geq 0.$$

Interpretation: The owner and planner differ because the owner puts full weight on sponsor profit and only indirect weight on user surplus. This is why the privately chosen certifier can be too friendly.

4.3 Proposition-level logic: baseline forum shopping

Baseline forum-shopping result

For a fixed concession, the sponsor chooses the most sponsor-friendly certifier that is still credible to users. If concessions are endogenous, weaker proposals make larger concessions and rely on tougher certification.

Economic message: Certification is a substitute for design concessions. A sponsor can make the proposal more attractive directly, or can choose a tougher certifier to make endorsement more credible.

4.4 Social regulation result

Private versus social choice

When the proposal is attractive, the sponsor chooses a certifier that is too complacent and makes too few concessions relative to the social optimum. When the proposal is weak, user participation binds tightly and regulation may not improve on the sponsor's choice. Regulating only certifier choice or only design can reduce welfare.

Interpretation: This is the paper's main welfare point. Users discipline the sponsor through their adoption constraint, but only partially. A strong proposal can exploit this discipline by choosing a friendly certifier and minimal concessions.

4.5 Property morphing

A more user-friendly certifier may demand design changes or concessions before endorsement. This creates property morphing: the forum can alter the proposal rather than merely certify it.

Property morphing

A certifier more user-friendly than the sponsor's preferred forum may bargain for more concessions. However, bargaining delays or failed negotiations can signal that the proposal is weak, limiting the certifier's ability to force changes.

Economic message: Certification institutions are not passive filters. They can shape product design, licensing terms, and standard-setting outcomes.

4.6 Sponsor downstream presence

If the sponsor is also a downstream user of the property, the sponsor internalizes some user surplus. The paper shows that stronger downstream presence can lead the sponsor to choose a less credible certifier, while still benefiting independent users.

Interpretation: The logic is that the sponsor's own user stake changes its design incentives. Because the sponsor already internalizes some user value, a friendly certifier can become less harmful to independent users.

4.7 Private information about user benefit

If the sponsor has private information about the distribution of b, forum choice can signal type. A pessimistic sponsor may choose a tougher or more informative forum, while an optimistic sponsor may pool or separate depending on beliefs and incentives.

Interpretation: The model adds adverse selection on top of moral-hazard-like forum shopping. The forum becomes both a certifier of the proposal and a signal of the sponsor's private information.

4.8 Certifier market power

If certifiers have market power, they may extract surplus from sponsors. The sponsor's outside options and the market structure for certifiers affect how much of the gain from certification is captured by certifiers versus sponsors.

Interpretation: Certifier market power changes the division of surplus but does not eliminate the basic credibility-versus-complacency tradeoff.

4.9 Multiple user categories

When multiple categories of users must coordinate, certification is a mechanism-design problem. A single certifier with appropriate weights on different user groups can implement the upper bound. Multiple certifiers can create coordination problems when adoption by all groups is required.

Multiple users

A centralized endorsement process can dominate multiple separate endorsements because it balances the participation constraints of different user groups. The owner makes the largest concessions to the users least eager to adopt.

Interpretation: This result is especially relevant for standards, platforms, and network goods, where one user group's adoption may be useless without the other groups.

4.10 Competing properties and strategic forum shopping

The final section introduces two rival sponsors. Each sponsor chooses a forum when users may adopt either property. Certifiers can be friendly or tough. Under competition, forum choice becomes strategic: a tough endorsement can differentiate one proposal from its rival, but may reduce the chance of adoption.

Competition result

Under competition, sponsors choose weakly less friendly, more credible certifiers than they would as monopolists. Depending on parameters, equilibrium behavior includes friendly submissions, tough submissions, differentiated submissions, or dual submissions.

Interpretation: Competition changes the value of credibility. A tough endorsement is costly because it reduces the probability of endorsement, but valuable because it differentiates the sponsor from its rival.

5 Identification and theoretical design

This is not an empirical identification paper. Its design is theoretical: it isolates how sponsor-chosen certifiers shape adoption and welfare.

5.1 What the model isolates

- **Certification as information.** Endorsement changes user beliefs about unknown quality b.
- **Certification as governance.** The certifier's bias α determines how much sponsor profit enters the endorsement decision.
- **Design as a substitute.** Concessions c make adoption more likely directly and make friendly certification more credible.
- **Forum shopping as strategic choice.** The sponsor chooses the certifier rather than treating it as exogenous.

Interpretation: The model identifies the core mechanism by holding the information structure simple and focusing on the sponsor's choice of certifier and concession.

5.2 Private versus social benchmark

The owner maximizes adoption probability times sponsor profit. The planner maximizes user surplus plus weighted sponsor profit. Comparing the two benchmarks identifies the welfare wedge.

$$\text{Private payoff} = [1 - F(b^*)]\pi(c),$$
$$\text{Social payoff} = E[X(b)(a + b + c + w\pi(c))].$$

Interpretation: The difference between these two objectives explains why an owner may choose a certifier that users would not choose on their own behalf.

5.3 Why certifier choice can be inefficient

The sponsor chooses the most friendly certifier consistent with user adoption. Users care only about expected utility conditional on endorsement. This leaves a margin: the owner can push credibility down to the participation constraint.

Economic message: The inefficiency comes from constrained persuasion. Users can reject noncredible endorsements, but they do not choose the certifier; the sponsor chooses the certifier subject to their willingness to adopt.

5.4 Why competition can improve credibility

A monopolist can select a friendly certifier as long as users accept. A rival sponsor changes the calculus: a tough certification can raise relative credibility and make users choose one proposal over another.

Interpretation: Competition does not necessarily make the market socially efficient, but it can push sponsors toward more informative certification than monopoly forum shopping would produce.

5.5 Main limitations of the theoretical design

- The model abstracts from repeated interactions and long-run certifier reputation building.
- The bias parameter α is treated as known and fixed, though real certifier governance may be opaque.
- The model uses simple binary endorsement, while real certifiers may issue grades, reports, or conditional approvals.
- Application costs and negotiation delays are largely simplified.
- Competition among multiple sponsors is modeled in a stylized two-type setting.

Interpretation: These simplifications make the core mechanism tractable. The paper is a first-pass theory of strategic certifier choice, not a full institutional model of all certification markets.

6 Figures and result interpretations

6.1 Figure 1: Competition and forum-shopping regimes

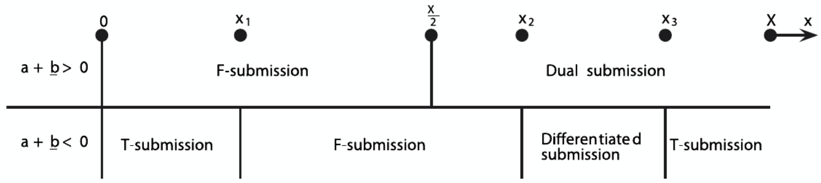

FIGURE 1

Figure 1. Original figure cropped from the paper. The horizontal axis is the probability parameter x relative to the mass X of favorable states. The top row corresponds to $a + \underline{b} > 0$; the bottom row corresponds to $a + \underline{b} < 0$. The regions show whether equilibrium forum shopping involves friendly submission, tough submission, differentiated submission, or dual submission.

Read:

- In the top case, $a + \underline{b} > 0$, users are more willing to adopt a weakly supported property. For low x, sponsors apply to friendly certifiers; for higher x, both use dual submission.
- In the bottom case, $a + \underline{b} < 0$, endorsement by a friendly certifier is less credible. The equilibrium moves from tough submission to friendly submission, then to differentiated submission, and finally back to tough submission as x rises.
- The key comparative static is that competition makes sponsors choose less friendly, more credible certification than a monopolist would.

Economic message: The figure summarizes the strategic forum-shopping game. Certification is no longer just about persuading users; it is also about differentiating one's proposal from a rival proposal.

The bottom row is especially important. Differentiated submission means one owner seeks a tough certifier while the other seeks a friendly certifier. This is a strategic differentiation outcome: certification choices function like product-positioning choices in a competitive market.

6.2 Notes-created Table: Key theoretical results and economic interpretation

Result	Formal content	Economic interpretation
Baseline forum shopping	Sponsor chooses the most friendly certifier satisfying $a + m(b^*) + c \geq 0$.	Certification credibility is pushed down to the user participation constraint.
Design choice	$\frac{1}{m(b^*)-b^*} + \frac{\pi'(c)}{\pi(c)} = 0$.	Concessions trade off adoption probability against sponsor profit.
Social wedge	Attractive proposals choose overly friendly certifiers and too few concessions.	Users cannot choose the forum; they only accept or reject its endorsement.
Property morphing	User-friendly certifiers may demand concessions.	Certifiers can alter property design, not just certify it.
Downstream presence	A sponsor who is also a user may choose a less credible certifier while independent users benefit.	Internalizing user value changes the meaning of certifier friendliness.
Multiple users	A single weighted certifier can implement the optimal mechanism.	Coordination across user categories favors centralized certification.
Competition	Rival sponsors apply to less friendly certifiers than monopolists.	Competition increases the value of credible differentiation.

Notes-created result table. This is not an original table from the paper. It summarizes the paper's main propositions and gives the economic reading used in these notes.

How it fits the paper: This table is paired with Figure 1 to make the visual section compact: Figure 1 covers the competitive regime map; the notes-created table summarizes the noncompetitive and extension results.

7 Short summary

- Sponsors use certifiers to persuade users about the value of ideas, standards, securities, or technologies.
- Certifiers vary in friendliness toward the sponsor relative to user interests.
- The sponsor chooses the most friendly certifier that remains credible to users.
- Weak proposals need tougher certifiers or greater concessions.
- Strong proposals tend to choose overly friendly certifiers from the social point of view.
- Regulating only certifier choice or only design can reduce welfare.
- Certifiers may morph property design by demanding concessions.
- Multiple user categories create a mechanism-design problem in which a centralized weighted certifier can dominate separate endorsements.
- Competition among rival sponsors makes credibility more valuable and pushes sponsors toward less friendly certification.
- The model applies to standard-setting organizations, journals, rating agencies, auditors, underwriters, venture capitalists, and other certifiers.

8 Conclusion

8.1 Core conclusion

Lerner and Tirole show that forum shopping is not simply opportunistic behavior in the informal sense. It is a predictable strategic response to the need for credible certification. Sponsors choose certifiers and concessions to maximize adoption while preserving profit. The chosen forum balances friendliness and credibility, but this balance can diverge from social welfare.

8.2 Economic intuition

A friendly certifier increases the chance of endorsement but weakens the meaning of endorsement. A tough certifier lowers the chance of endorsement but makes adoption more likely conditional on endorsement. The sponsor wants the friendly side of this tradeoff; users want credible information. The equilibrium certifier is the friendliest one that users still trust.

8.3 Incremental lesson

The paper's distinctive lesson is that certification markets are endogenous institutional markets. The quality of certification cannot be evaluated independently of who chooses the certifier and what concessions the sponsor can make. Certification is therefore both an information problem and a mechanism-design problem.

8.4 Policy implication

A regulator who wants to improve certification cannot simply mandate a tougher forum or a better design in isolation. The sponsor may adjust the other dimension in response, and partial regulation can lower welfare. Good policy must consider certifier choice, design concessions, user participation, and competitive context jointly.

8.5 Applications

The model sheds light on standard-setting organizations, academic journals, rating agencies, auditors, investment banks, venture capitalists, and product-review institutions. In all of these settings, the sponsor cares about endorsement and may choose the forum strategically.

8.6 Limitations and future work

The model is stylized. It abstracts from rich grading systems, repeated certifier reputation, entry and exit in certification markets, endogenous certifier governance, bargaining delays, and complex multi-sponsor standard-setting dynamics. These are natural extensions of the framework.

8.7 Takeaway

The key takeaway is that certifier credibility is endogenous. Users should ask not only what a certifier says, but why that certifier was chosen. The sponsor's forum-shopping decision is itself part of the information conveyed by certification.